

# 02\_DUMBASS\_UNIVERSAL\_MEMORY\_SPEC.md

## The Database & Universal Memory Bank Across Split Services (#d.u.m.b.a.s.s.)

### 1. Document Authority & Supersession

This specification establishes the official architecture for the **#d.u.m.b.a.s.s.** memory subsystem across mobile edge devices, local servers, and cloud instances. It envelops, unifies, and supersedes:

- **\*\*SQLite.txt**
- **Attaching #dumbass and #sop-Xi**
- **Continual harness online adaptation for self-improving foundation agents.txt**
- **The Global Information Layer & Storage Matrix.txt**
- **Mem0 architecture notes (Folder 4)**

### 2. Core Operational Law: The Integrated Memory Engine

|                                  |                                                 |                            |
|----------------------------------|-------------------------------------------------|----------------------------|
| #d.u.m.b.a.s.s. MEMORY SUBSYSTEM |                                                 |                            |
| Component                        | Architectural Responsibility                    | Hardware & Network Binding |
| mem0                             | In-session episodic state & user habits         | Node Alpha (Mobile RAM)    |
| OB1 Protocol                     | Ground-truth static retrieval protocols via MCP | Ubiquitous Network-Wide    |

| OmniRoute Gateway | Token conservation, prompt caching & cost route | Localhost Port 20128 |

| Reasoning Bank | Multi-model execution ledger & crash recovery | JSON / KV Store |

| SQLite | Embedded zero-latency relational/KV tables | Node Alpha Flash / Disk |

| PostgreSQL | Authoritative relational & vector persistence | Node Beta (Jetson Server) |

| Continual Harness | Online reset-free self-improvement loop | Supervisory State Machine |

| | |

## Beginner-Proof Explanation: How the Layers Work Together

1. **mem0**: Remembers what you said 2 minutes ago and how you like your answers structured. It maintains short-term personalization so the model doesn't ask you the same questions repeatedly.
2. **OB1 (Open Brain Protocol)**: When the model needs to reference a 500-page Qualcomm manual or system schema, it queries OB1 over MCP. OB1 extracts the exact grounded paragraphs so the model never has to guess or hallucinate.
3. **OmniRoute**: Sits on <http://localhost:20128/v1>. When an agent makes an API call, OmniRoute checks if the prompt was already cached, routes cheap queries to local weights, and sends heavy queries to cloud providers while minimizing token spend.
4. **Reasoning Bank ([active\\_execution\\_paths.json](#))**: When an agent undertakes a 10-step programming task, it writes each step to this ledger. If the battery dies or the process is killed at Step 6, the system reads this ledger upon reboot and resumes at Step 7 without losing state.
5. **SQLite**: The local database on your phone. Requires zero setup, zero network connection, and answers queries in under 5 milliseconds.
6. **PostgreSQL**: The master database on your Jetson server that backs up all history, code graphs, and enterprise records across the network.
7. **Continual Harness**: Allows agents to learn and refine prompt instructions during live execution. If an update causes an error, it immediately triggers an automated snapshot rollback. The base system prompt is never modified.

### 3. Universal JSONL Marker Schema

Every asset indexed by [#dumbass](#) receives an atomic JSONL marker record matching this schema:

```
{  
  
  "record_id": "MEM_A102_20260904_SCHEMA",  
  
  "document_path": "02_wiki_md/architectures/npu_topology.md",  
  
  "tier": 2,  
  
  "category": "TECHNICAL_REFERENCE",  
  
  "metadata": {  
  
    "title": "NPU Memory Topology & FastRPC Bindings",  
  
    "description": "Zero-copy shared memory allocation for Snapdragon 8 Elite Hexagon NPU.",  
  
    "content_hash": "e3b0c44298fc1c14",  
  
    "file_size_bytes": 4820,  
  
    "last_modified": "2026-09-04T07:45:00Z"  
  
  },  
  
  "retrieval_tokens": ["qualcomm", "qnn", "fastrpc", "asharedmemory", "dma_buf", "htp"],  
  
  "entry_points": {  
  
    "mcp_method": "knowledge.retrieve",  
  
    "cli_command": "/skill run npu-topology"  
  
  }  
  
}
```

---

## 4. Git Orchestration & Cross-Repo Sync Hook

To ensure `#dumbass` (`novae-xorpus`) and consumer repositories (like `aesop-xi`) stay synchronized across sessions without duplication, use this automated Git post-commit hook:

```
#!/usr/bin/env bash

#
=====

# Git Post-Commit Synchronization Hook for #dumbass

# Location: ~/repos/aesop-xi/.git/hooks/post-commit

#
=====

set -euo pipefail

VAULT_DIR="${HOME}/novae-xorpus"

CONSUMER_DIR="${HOME}/repos/aesop-xi"

echo "[*] Triggering #dumbass master ledger regeneration..."

# 1. Ensure symlinks exist

mkdir -p "${VAULT_DIR}/projects/aesop-xi"

ln -sf "${CONSUMER_DIR}/RESUME.md" "${VAULT_DIR}/projects/aesop-xi/RESUME.md"

ln -sf "${CONSUMER_DIR}/CLAUDE.md" "${VAULT_DIR}/projects/aesop-xi/CLAUDE.md"

# 2. Execute master regeneration utility

if [ -f "${VAULT_DIR}/tools/regenerate_masters.sh" ]; then

    bash "${VAULT_DIR}/tools/regenerate_masters.sh"

    echo "[✓] MASTER-RESUME.md and MASTER-CLAUDE.md synchronized successfully."
```

